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Studierichting: Informatica

Oefeningenreeks 4A nr. 14.4

$$\begin{aligned} & \int \ln^2 x dx \\ &= x \ln^2 x - \int x d(\ln^2 x) = x \ln^2 x - \int x \cdot 2 \ln x \cdot \frac{1}{x} dx \\ &= x \ln^2 x - 2 \int \ln x dx = x \ln^2 x - 2x \ln x + 2 \int x d(\ln x) \\ &= x \ln^2 x - 2x \ln x + 2x + c = x (\ln^2 x - 2 \ln x + 2) + c \end{aligned}$$

References